

AZAM MOHAMED

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EDUCATION

University of California, Santa Cruz, Santa Cruz, CA

Sep 2023 – Aug 2026

B.S. Computer Engineering, Minors in Electrical Engineering and Computer Science

Relevant Coursework: Operating Systems, Computer Systems, Data Structures, Computer Architecture, Embedded Systems

WORK EXPERIENCE

UC Santa Cruz jLab (Josephson Lab), *Undergraduate Researcher, ENTS Project*, Santa Cruz, CA

Jul 2026 – Present

- Trace the full ENTS data path, firmware to transport to the DirtViz dashboard, to locate where field readings are lost or silently corrupted before display.
- Decode nanopb Protocol Buffer telemetry to verify packets survive transport intact, isolating failures to a specific stage rather than the pipeline at large.
- Build a repeatable validation procedure so any lab member can certify a node's data end to end, replacing ad hoc spot checks that let failures reach production dashboards.

UC Santa Cruz, *Undergraduate Researcher, Independent Study (CSE 198)*, Santa Cruz, CA

Jun 2026 – Present

- Configuring and testing the department's PICO compute cluster for Kubernetes research workloads under Prof. Abel Souza.
- Evaluating Incus system containers and VMs on Linux to build reproducible, isolated experiment environments, prototyping container and VM provisioning workflows.

Santa Cruz Beach Boardwalk, *Rides Operator*, Santa Cruz, CA

Feb 2026 – Jun 2026

- Ensured safe operation of mechanical systems while enforcing safety protocols for 100+ daily guests, building practical intuition for fault detection in safety-critical systems.

PROJECTS

Pintos Kernel Scheduler

C, OS Internals, Thread Scheduling

- Reworked Pintos thread sleeping to use an ordered sleep_list with a custom wakeup_less comparator, replacing busy-waiting with efficient timed wakeups inside the kernel.
- Woke sleeping threads from the 8254 timer interrupt handler by popping due threads from the sorted list front, disabling interrupts around insertion to avoid races with the ISR; passed all six alarm tests under QEMU.

Thread-Safe Binary Search Tree

C, pthreads, Mutexes, Valgrind

- Implemented a thread-safe binary search tree in C using fine-grained mutex locking to support concurrent insert, search, and delete operations across multiple threads.
- Hardened it against races and leaks using Valgrind and multi-threaded stress tests, reaching clean runs with zero reported memory errors under sustained contention.

Multi-Client TCP Chat Server

C, POSIX Sockets, select()

- Built a TCP chat server in C using select()-based I/O multiplexing to serve many clients concurrently on a single thread, handling connects, disconnects, broadcast, and clean session teardown without disrupting other users.

Autonomous Competition Robot ("Sluggers of the Lost Goal")

Embedded C, ESP-IDF, UART, HSM, Real-Time Systems

- Owned the ESP32-C3 subsystem on a dual-microcontroller robot (ESP32-C3 + PIC32), defining a single-byte UART protocol that let teammates develop the driving state machine independently against a stable interface.
- Took ownership of the failing navigation hierarchical state machine near checkoff, diagnosing faults across state logic, sensor polarity, and motor direction, turning an unreliable run into a robot that consistently completed the full field sequence.

2x2 Matrix Solver ASIC (TinyTapeout, SKY130)

Verilog, cocotb, OpenLane, Fixed-Point

- Designed a sequential 2x2 linear-system solver ($Ax=b$, Q4.4 fixed point) in Verilog for the TinyTapeout SKY130 flow, fitting the datapath and a 12-state control FSM in a 1x1 tile targeting 50 MHz.
- Verified via cocotb through RTL and gate-level simulation and hardened the design to GDS with the OpenLane-based TinyTapeout toolchain.

SKILLS AND INTERESTS

Languages: C, C++, Python, MATLAB, Verilog, Assembly (x86/RISC-V)

Systems: Linux, Operating Systems, Systems Programming, Concurrency, POSIX Threads (pthreads), Sockets, select() I/O Multiplexing, Memory Management, Client-Server Architecture

Tools: Git, GDB, Valgrind, Docker, Vivado, cocotb, SQLite, WSL, SSH, Neovim, tmux

Embedded: ESP-IDF, FreeRTOS, I2C, SPI, UART, PWM, BLE

Activities: Member, UC Santa Cruz High Performance Computing Club

Interests: Cricket, Table Tennis, Tennis, Beach Volleyball, Cooking